LECTURE 05

DOMAIN OF ALGEBRAIC FUNCTIONS

Polynomial:

Sum of many monomial is called polynomial.

An expression of the form cx^n, where c is any scalar and n is non-negative integer.

Any non-zero real number is called scalar.

$$y = x^2$$

Domain: $\{x|x\epsilon\mathcal{R}\}$

CONT…

Non-Polynomial:

Case 1: Involving denominator

$$y = \frac{1}{x}, \qquad x \neq 0$$

Domain: $\{x | x\epsilon\mathcal{R} \wedge x \neq 0\}$ $or\,(-\infty, 0) \cup (0, \infty)$

$$y = \frac{1}{x+1}$$

$$x + 1 \neq 0$$

$$x \neq -1$$

Domain: $\{x | x\epsilon\mathcal{R} \wedge x \neq -1\}$

CONT…

$$y = \frac{1}{(x-1)(x+1)}$$

$$\Rightarrow (x-1)(x+1) \neq 0$$

$$(x-1) \neq 0, \qquad (x+1) \neq 0$$

$$x \neq 1, \qquad x \neq -1$$

$$\{x | x\epsilon\Re \wedge x \neq \pm 1\}$$

Interval: $(-\infty, -1) \cup (-1,1) \cup (1, \infty)$.

Case II: Radical sign involve

$$y = \sqrt{x}\,, \qquad x \geq 0$$

Domain: $\{x | x\epsilon\mathcal{R} \wedge x \geq 0\}$, Interval: $[0, \infty)$

CONT…

$$y = \sqrt{x+1},$$

$$x + 1 \geq 0$$

$$x \geq -1$$

$$\{x | x \epsilon \mathcal{R} \wedge x \geq -1\}$$

$$y = \sqrt{(x+1)(x-1)}$$

$$(x-1)(x+1) \geq 0$$

Interval: $(-\infty, -1] \cup [1, \infty)$

CONT…

Case III: Radical in denominator

$$y = \frac{1}{\sqrt{x}}, \qquad x > 0$$

Domain: $\{x|x\epsilon\mathcal{R} \wedge x > 0\}$

$$y = \frac{1}{\sqrt{2x - 3}}$$

$$2x - 3 > 0$$

$$\mathrm{x} > 3/2$$

Domain: $\{x|x\epsilon\mathcal{R} \wedge x > 3/2\}$

EXPONENTIAL FUNCTION

$$y = e^{f(x)}$$

where, $f(x)$ is polynomial.

$$y = e^{x}, e^{x^2}, e^{x^3}, e^{3x^2+2x+1}$$

Domain: $\{x|x\epsilon\mathcal{R}\}$

$$y = e^{1/x}$$

where, $1/x$ non-polynomial

$$\{x|x\epsilon\mathcal{R} \wedge x \neq 0\}$$

LOGARITHMIC FUNCTION

$$y = lnx, \qquad x > 0$$

$$y = lnx^2, \qquad x^2 > 0$$

$$y = \ln(x-1)\,(x+1)$$

$$(x-1)(x+1) > 0$$

$$y = \ln(f(x)),$$

$$f(x) > 0$$

In Exercises 37–60 sketch the graph and determine the domain and range of f.

37 $f(x) = -4x + 3$

38 $f(x) = 4x - 3$

39 $f(x) = -3$

40 $f(x) = 3$

41 $f(x) = 4 - x^2$

42 $f(x) = -(4 + x^2)$

43 $f(x) = \sqrt{4 - x^2}$

44 $f(x) = \sqrt{x^2 - 4}$

45 $f(x) = \dfrac{1}{x - 4}$

46 $f(x) = \dfrac{1}{4 - x}$

47 $f(x) = \dfrac{1}{(x - 4)^2}$

48 $f(x) = -\dfrac{1}{(x - 4)^2}$

49 $f(x) = |x - 4|$

50 $f(x) = |x| - 4$

51 $f(x) = \dfrac{x}{|x|}$

52 $f(x) = x + |x|$

53 $f(x) = \sqrt{4 - x}$

54 $f(x) = 2 - \sqrt{x}$